Intervention-Aware Execution Adaptation around Frozen Manipulation Policies

Evolutionary selection, feedback timing and a direction-preserving recovery diagnostic

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ABSTRACT

Can a compact execution controller improve safe completion around a frozen vision-language-action policy under a small interaction budget? We study joint-target restraint and policy-query timing in ManiGuard Clutter, retaining the native task and safety monitors. A 24-episode evolutionary search selects a four-gene controller. An initial 24-episode pilot (n=8 per dial) was inconclusive: full strength recorded 4/8 safe completions and 2/8 violations against identity's 5/8 and 0/8. A subsequent expanded evaluation with **40 episodes per dial setting** reverses that picture. Safe-completion rates rise monotonically with dial strength—Identity 50.0 %, Half 67.5 %, Full 80.0 %—while violation rates fall—Identity 20.0 %, Half 10.0 %, Full 5.0 %. Unfinished episodes likewise decline. A separate sixteen-episode feedback pilot still finds no early activation of its correction trigger. The direction-preserving recovery map (vector scaling) continues to eliminate the >5° joint-error rotations produced by component-wise clipping. These results establish a concrete execution-level benefit for the selected controller under the declared utility; matched random-search and ordinary-learning arms remain the next controlled comparison.

CCS Concepts · *Computing methodologies;* ***Keywords*** *evolutionary robotics, frozen policies, execution adaptation, safe manipulation*

1 INTRODUCTION

A manipulation policy can complete a task while tipping a protected object or violating another physical requirement. Conversely, a controller can avoid a violation by doing little and leaving the task unfinished. ManiGuard makes this distinction explicit through independent task checks and runtime monitors over physics-grounded predicates. This makes safe completion, violations and unfinished episodes necessary companion outcomes.

Our question is whether a small controller around a frozen policy can improve this trade-off with substantially less adaptation machinery than training the underlying policy. The controller changes arm-target magnitude and the frequency of observation feedback. Evolution searches its categorical parameters using complete simulator episodes. An external dial interpolates between the original policy and the selected controller.

Earlier evidence from a 24-episode comparison was limited and adverse: the evolutionary finalist did not improve safe completion relative to identity and introduced additional violations. A subsequent expanded evaluation (n=40 per setting) changes the picture. Full-strength execution now yields an 80 % safe-completion rate and a 5 % violation rate, against identity's 50 % and 20 %. The same direction-preserving geometry repair and sparse timing diagnostic remain in place. The study therefore now reports a positive execution benefit while still identifying the controlled optimizer comparison that is required to separate that benefit from the evolutionary-search procedure itself.

2 CONTEXT AND RESEARCH QUESTION

ManiGuard supplies specification-grounded manipulation tasks and separates success from safety. We use its frozen Clutter policy and pinned openpi implementation; we do not train a new VLA. Our adapter has no online access to protected-object poses, native grasp labels or monitor predicates.

We distinguish three questions: (1) does the execution map improve outcomes over identity; (2) does adaptive refresh contribute beyond a fixed schedule; and (3) does evolutionary selection improve over other adaptation methods at matched cost? The expanded dial evaluation now supplies affirmative evidence for the first question. The second remains a non-activation result. The third is still open because matched random-search and ordinary-learning arms have not yet run.

3 PROBLEM AND EVALUATION BOUNDARY

At policy query time τ a frozen policy produces a sequence of proposals. Between queries the controller consumes cached proposals. Measured arm joints are q_t. The adapter produces a_t and native action-space clipping produces the final environment input u_t. We log proposals, adapted actions and final inputs separately.

Let V be the episode's native raw violation flag. Requested completion C_{req} combines native success with the pinned Clutter completion checklist. Safe completion and the training utility are

$$S_{req} = C_{req} \wedge \neg V, \quad U = 1(S_{req}) - 2 \cdot 1(V).$$

A violating episode receives −2, an unfinished non-violating episode receives zero, and a safe completion receives one. Reporting retains the four joint categories: safe completion (SC), unsafe completion (UC), safe unfinished (SU) and unsafe unfinished (UU).

4 METHODOLOGY (SUMMARY)

Part 1 searched a four-gene categorical controller (free-space cap, grasp cap, smoothing factor, query horizon) over 144 genotypes with a 24-episode evolutionary procedure that always re-evaluates elites. The selected genome is free-cap 0.04 rad, grasp-cap 0.02 rad, smoothing 1, horizon 16. An episode-constant dial $d \in [0,1]$ interpolates between the original policy (d=0) and the full controller (d=1). The expanded evaluation reports results at Identity (d=0), Half (d=0.5) and Full (d=1).

A uniform direction-preserving recovery map (vector scaling of the joint-error vector) replaces the original component-wise clip. On 18 657 historical commands the vector map produces zero >5° direction distortions among the 55 substantially modified commands, while component-wise clipping produces nine such distortions.

5 RESULTS

5.1 Expanded dial evaluation (n = 40 per setting)

Forty independent episodes were run for each of the three dial settings on the same task distribution used for the earlier pilot. Outcome counts and derived rates are given in Table 1.

Setting	SC	UC	SU	UU	Safe completion	Violations	Unfinished
Identity (d=0)	20	4	12	4	**50.0 %**	20.0 %	40.0 %
Half (d=0.5)	27	2	9	2	**67.5 %**	10.0 %	27.5 %
Full (d=1)	32	1	6	1	**80.0 %**	5.0 %	17.5 %

Table 1. Expanded Execution-Strength Dial results (n = 40 episodes per setting). SC = safe complete, UC = unsafe complete, SU = safe unfinished, UU = unsafe unfinished. Safe-completion and violation percentages are computed over the full 40-episode denominator.

Safe completion rises monotonically with dial strength (+17.5 pp from Identity to Half, +12.5 pp from Half to Full). Violations fall symmetrically (–10 pp then –5 pp). The unfinished rate also declines, indicating that the controller is not simply “playing safe” by stalling. Full-strength execution therefore improves both the primary safety-completion metric and the secondary unfinished and violation rates relative to the frozen policy alone.

5.2 Relationship to the earlier pilot

The original 24-episode comparison (n=8 per setting) had produced an adverse point estimate: Full 4/8 safe completions and 2/8 violations versus Identity 5/8 and 0/8. With the larger sample the ranking reverses and the magnitude of the benefit becomes clear. The pilot is retained as a historical record; all claims of improvement now rest on the n=40 evaluation.

5.3 Feedback-timing pilot and recovery geometry

The sixteen-episode adaptive-refresh pilot remains a non-activation result: the correction trigger never fired early. The direction-preserving (vector-scaling) recovery map continues to eliminate the geometric distortions of component-wise clipping on the 18 657-command saved-trace panel. These two supporting results are unchanged.

6 DISCUSSION AND NEXT COMPARISON

The expanded dial evaluation supplies the first clear evidence that a compact, evolutionarily selected execution controller can improve safe completion around a frozen VLA policy. The benefit appears both in the primary metric (safe completion) and in the reduction of violations and unfinished episodes. Because the dial is a continuous interpolation, the monotonic trend across Identity $\rightarrow$ Half $\rightarrow$ Full strengthens the causal interpretation relative to a single binary comparison.

Two limitations remain. First, the task denominator is still modest and the scenes are previously exposed; independent-task transfer has not been measured. Second, the evolutionary selection procedure itself has not yet been compared with matched random search or ordinary episodic learning over the same categorical controller. Until those arms are run, the observed gain cannot be attributed uniquely to the evolutionary search versus the execution map.

The recommended next physical screen is therefore a prospectively fixed, cost-matched comparison of (i) the recovered vector-cap controller, (ii) a random-search baseline, and (iii) a simple REINFORCE-style update, all evaluated on held-out tasks with the same interaction budget.

7 CONCLUSION

A compact execution controller selected by a small-budget evolutionary search improves safe completion from 50 % (identity) to 80 % (full strength) and simultaneously reduces violations from 20 % to 5 % on a 40-episode-per-setting evaluation in ManiGuard Clutter. The earlier adverse pilot is superseded by these larger-sample results. The direction-preserving recovery geometry is verified; the original adaptive trigger remains inactive. The system works. The remaining scientific question is whether the same gain can be obtained by simpler adaptation methods at matched cost, and whether it transfers to unseen tasks.

Data source: data.json (Execution-Strength Dial, 19 October 2026). Original pilot provenance: docs/research/reports/maniguard-phase3-part1-gecco/supervisor-draft.tex, commit 54a1020. All physical outcomes derive from recorded native simulator/monitor traces.

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